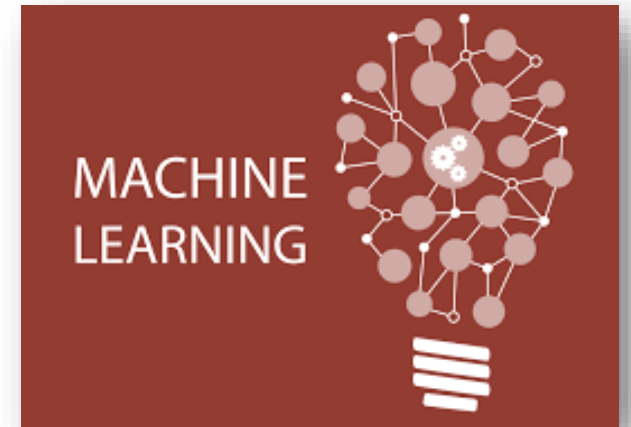




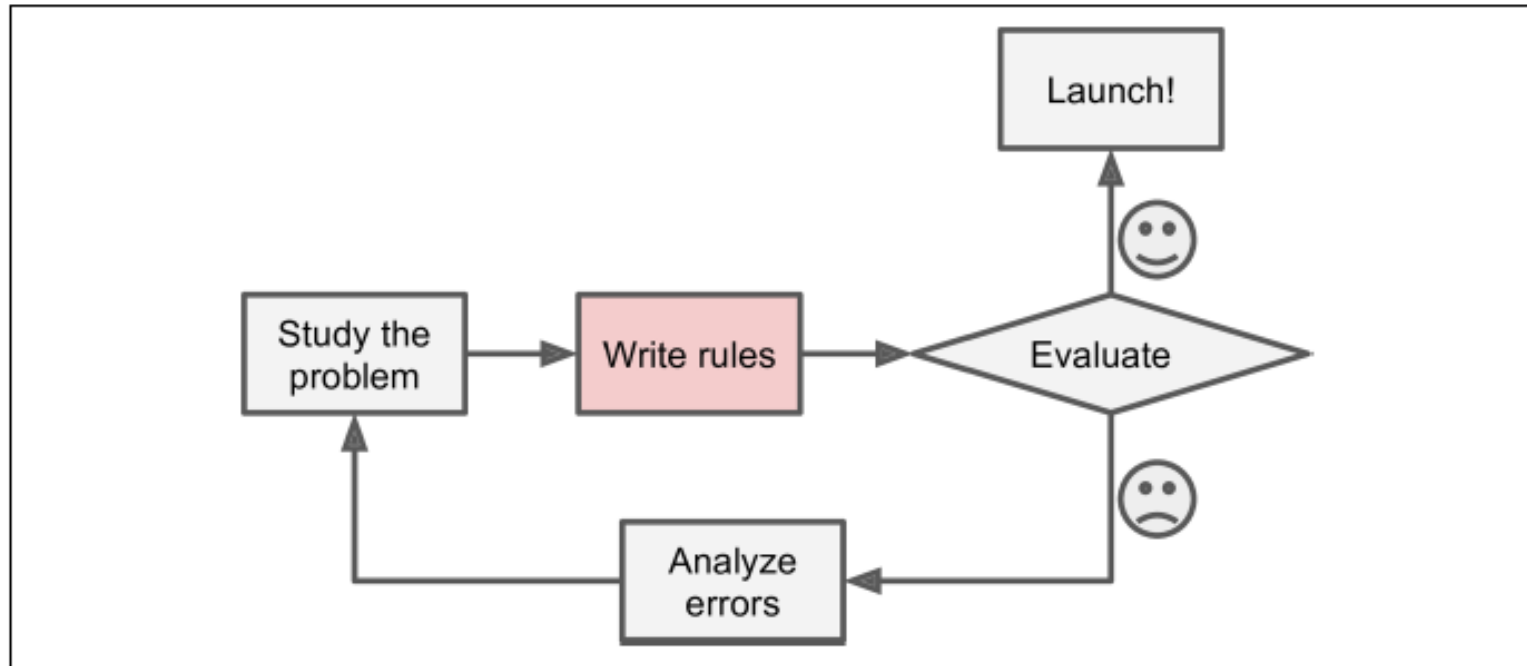
Machine Learning

DR.SATTAM ALMATARNEH





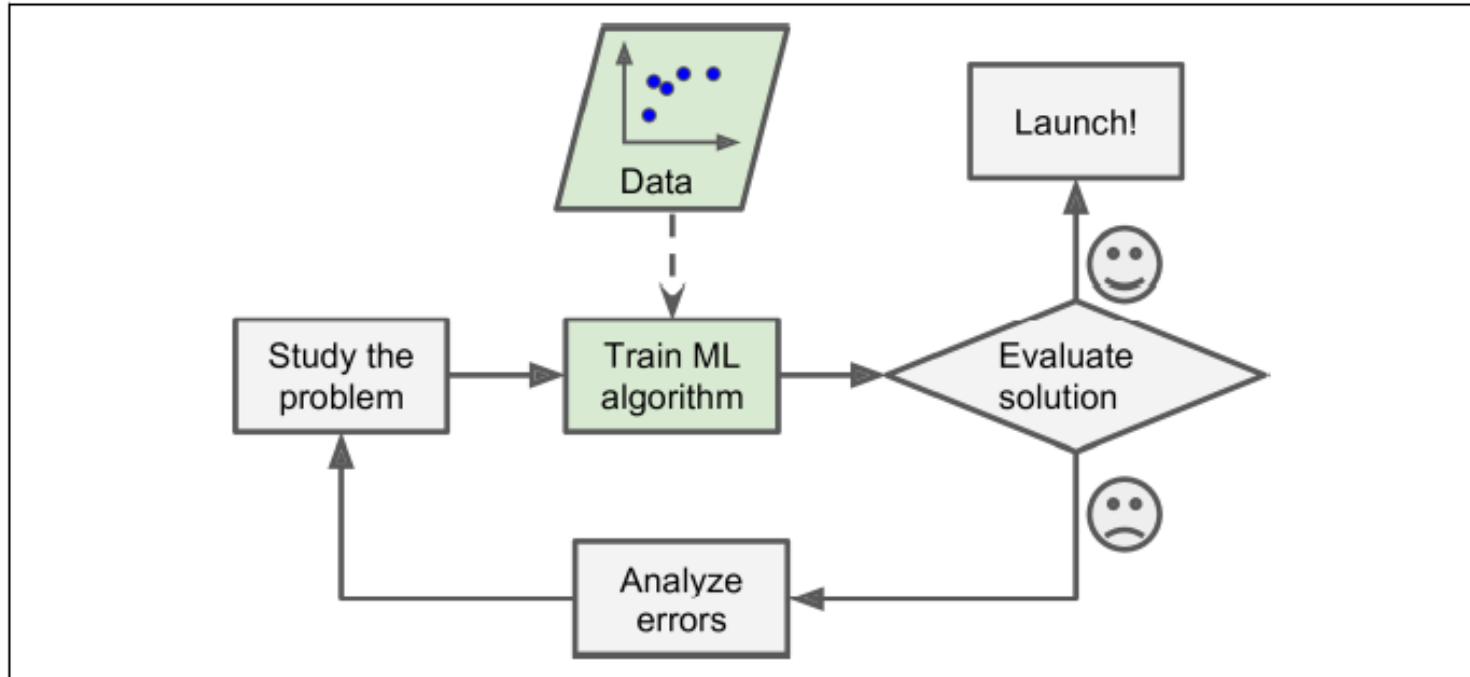
Machine Learning approach vs traditional approaches



The traditional approach



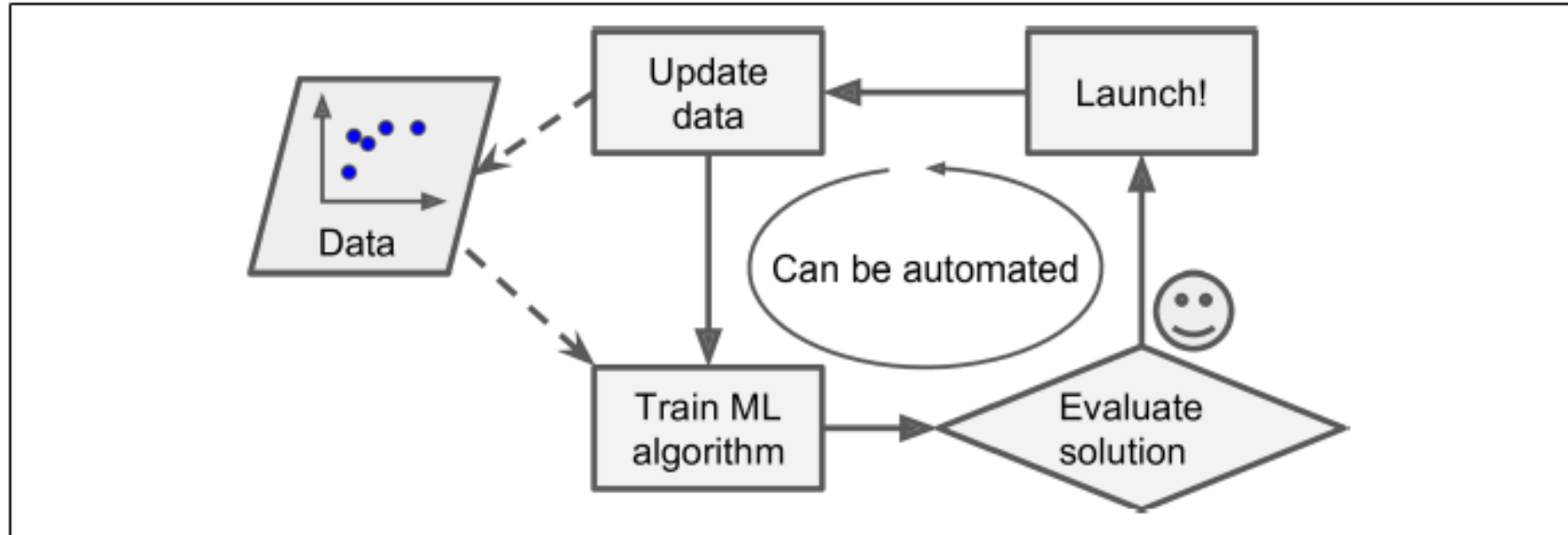
Machine Learning approach vs traditional approaches



Machine Learning Approach



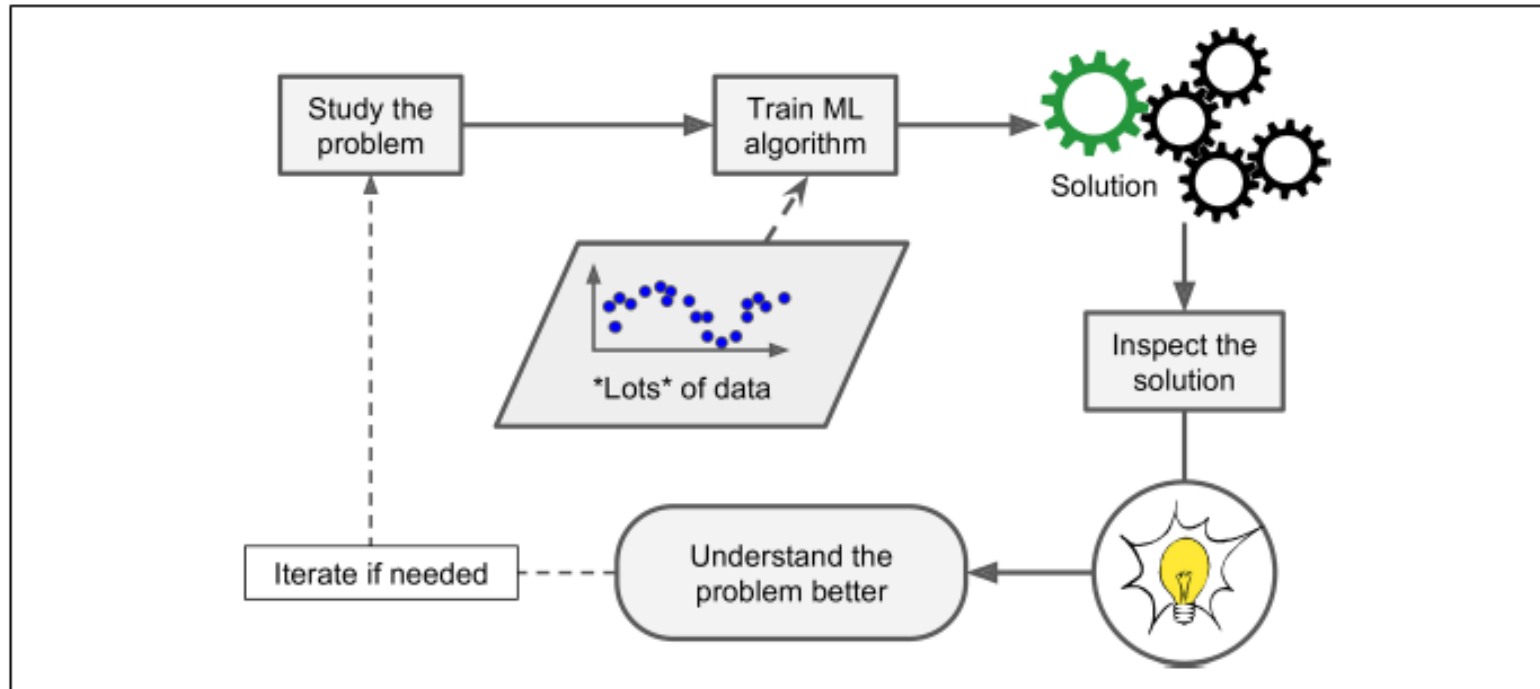
Machine Learning approach vs traditional approaches



Automatically adapting to change



Machine Learning approach vs traditional approaches



Machine learning can help humans learn



Machine Learning is great for:

- Problems for which existing solutions require a lot of hand-tuning or long lists of Rules: one machine learning algorithm can often simplify code and perform better.
- Complex problems for which there is no good solution at all using a traditional Approach: the best machine learning techniques can find a solution.
- Fluctuating environments: a machine learning system can adapt to new data.
- Getting insights about complex problems and large amounts of data.



Types of Machine Learning Systems

There are so many different types of Machine Learning systems that it is useful to classify them in broad categories based on:

- Whether or not they are trained with human supervision (supervised, unsupervised, semisupervised, and Reinforcement Learning)
- Whether or not they can learn incrementally on the fly (online versus batch learning)
- Whether they work by simply comparing new data points to known data points, or instead detect patterns in the training data and build a predictive model, much like scientists do (instance-based versus model-based learning)

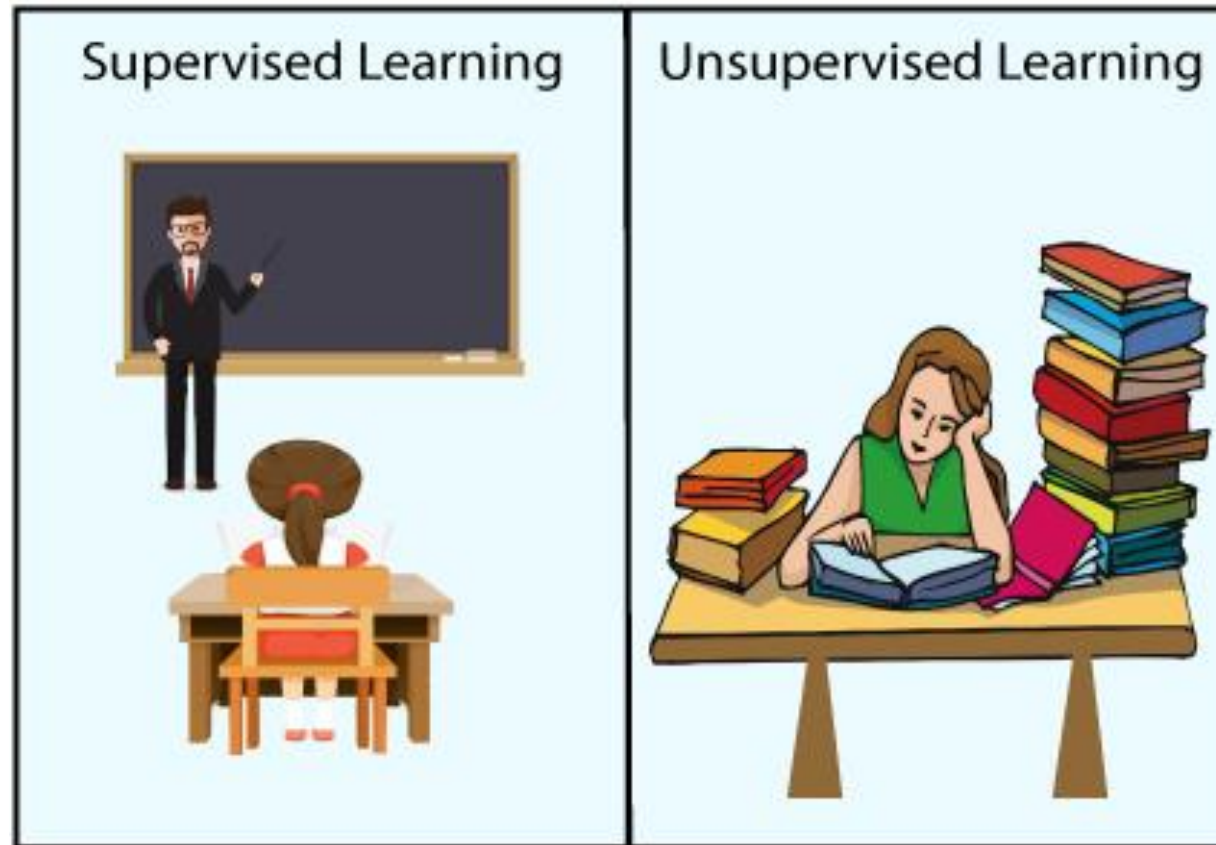


Types of Learning

- **Supervised (inductive) learning**
 - Given: training data + desired outputs (labels)
- **Unsupervised learning**
 - Given: training data (without desired outputs)
- **Semi-supervised learning**
 - Given: training data + a few desired outputs
- **Reinforcement learning**
 - Rewards from sequence of actions



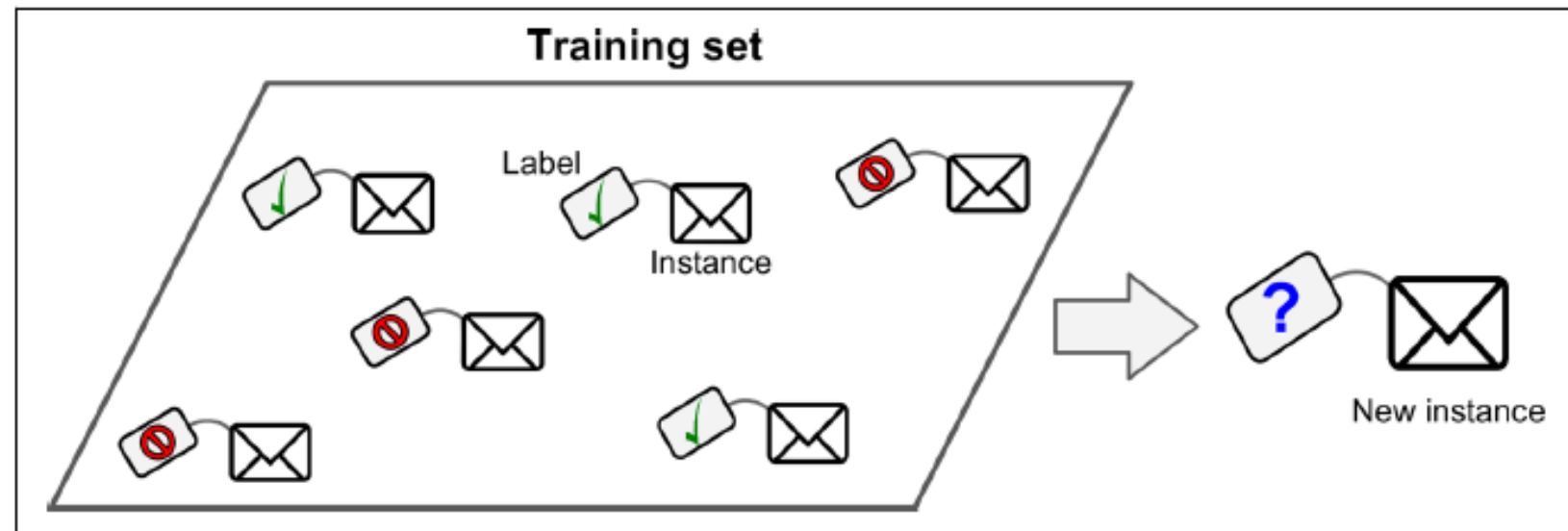
Supervised/Unsupervised Learning





Supervised/Unsupervised Learning

In *supervised learning*, the training data you feed to the algorithm includes the desired solutions, called *labels*

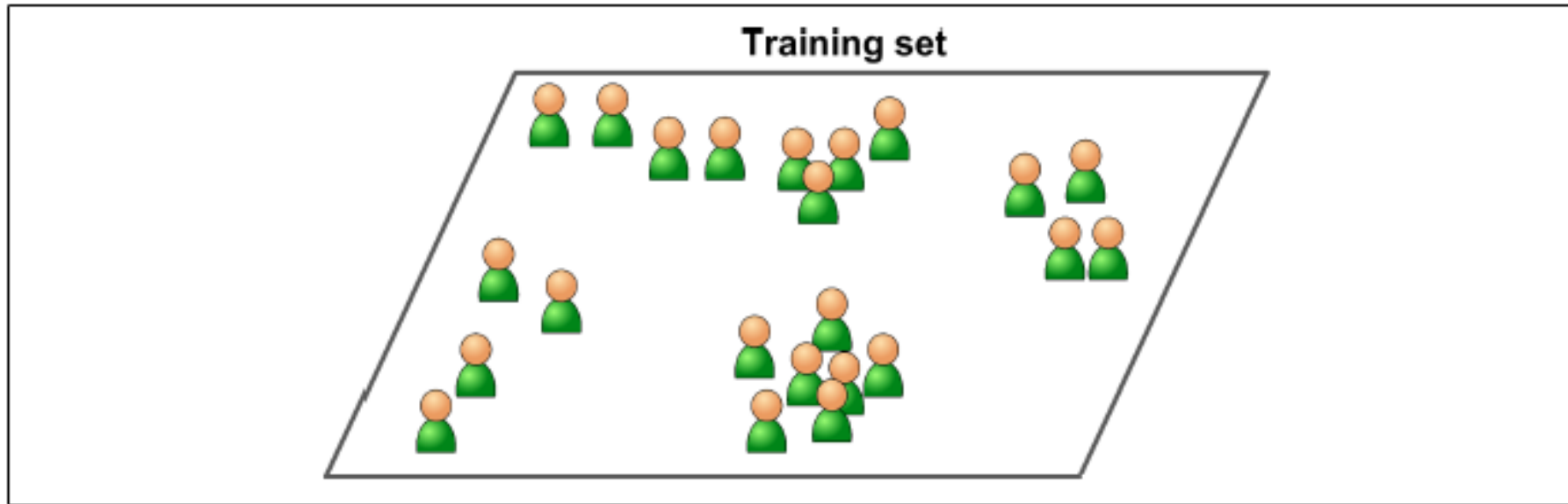


Supervised learning



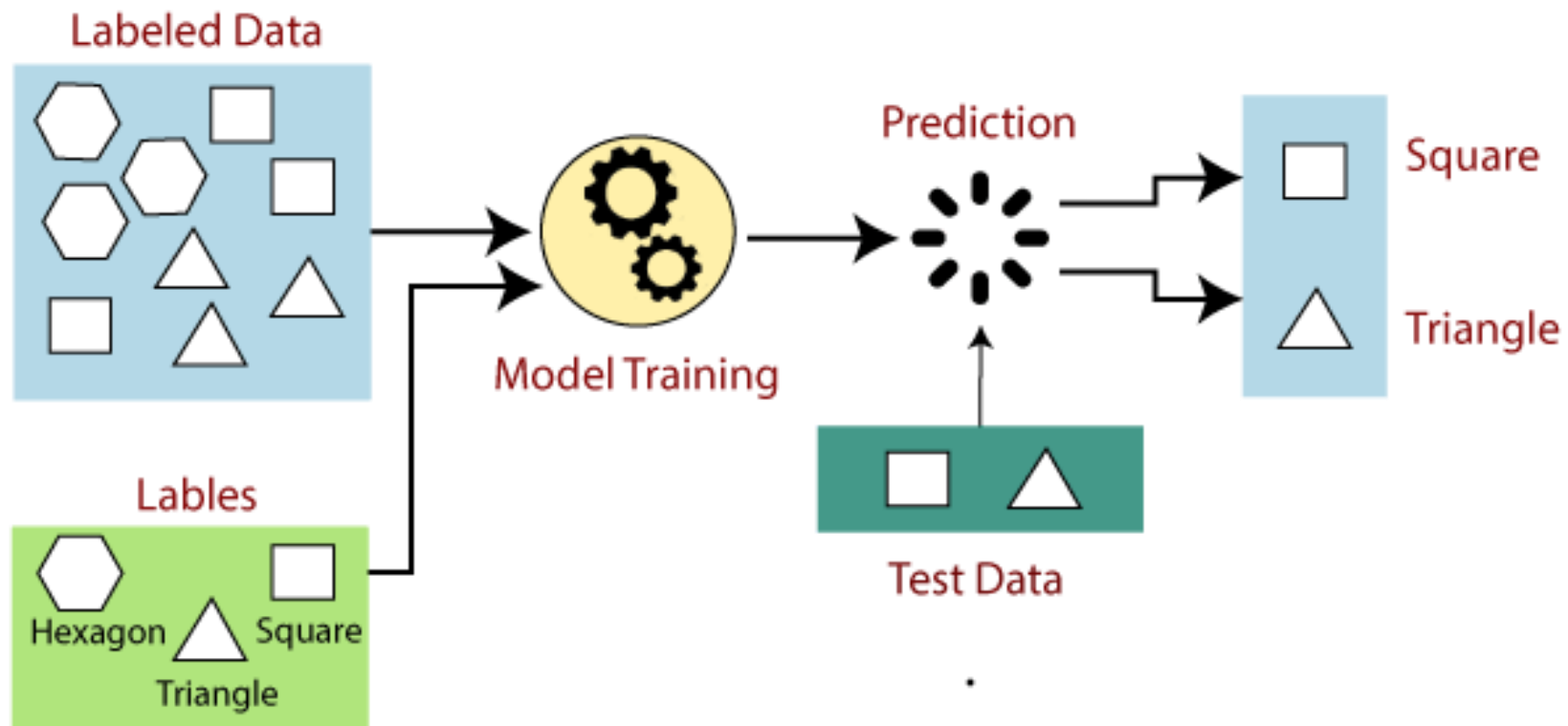
Supervised/Unsupervised Learning

In *unsupervised learning*, as you might guess, the training data is unlabeled
The system tries to learn without a teacher.



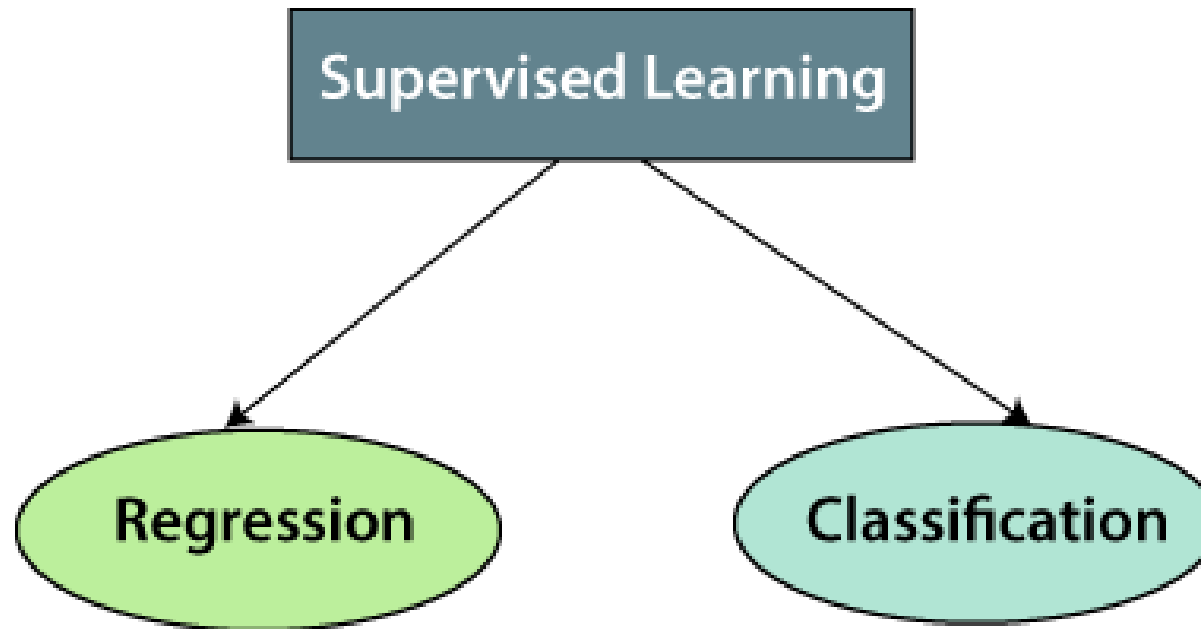


How Supervised Learning Works?





Types of supervised Machine learning Algorithms:





Regression

Regression algorithms are used if there is a relationship between the input variable and the output variable. It is used for the prediction of continuous variables, such as weather forecasting, market trends, etc. Below are some popular regression algorithms which come under supervised learning:

- Linear regression
- Regression trees
- Non-linear regression
- Bayesian linear regression
- Polynomial regression



Classification

Classification algorithms are used when the output variable is categorical, which means there are two classes such as yes-no, male-female, true-false, etc.

random forest
decision trees
logistic regression
support vector machines



Advantages of Supervised learning:

- With the help of supervised learning, the model can predict the output on the basis of prior experiences.
- In supervised learning, we can have an exact idea about the classes of objects.
- Supervised learning model helps us to solve various real-world problems such as **fraud detection, spam filtering**, etc.



Disadvantages of supervised learning:

- Supervised learning models are not suitable for handling the complex tasks.
- Supervised learning cannot predict the correct output if the test data is different from the training dataset.
- Training required lots of computation times.
- In supervised learning, we need enough knowledge about the classes of object.



Unsupervised Machine Learning

Unsupervised learning is a type of machine learning in which models are trained Using unlabeled dataset and are allowed to act on that data without any supervision.



Unsupervised Machine Learning

Unsupervised learning cannot be directly applied to a regression or classification problem because unlike supervised learning, we have the input data but no corresponding output data. The goal of unsupervised learning is to **find the underlying structure of dataset, group that data according to similarities, and represent that dataset in a compressed format.**



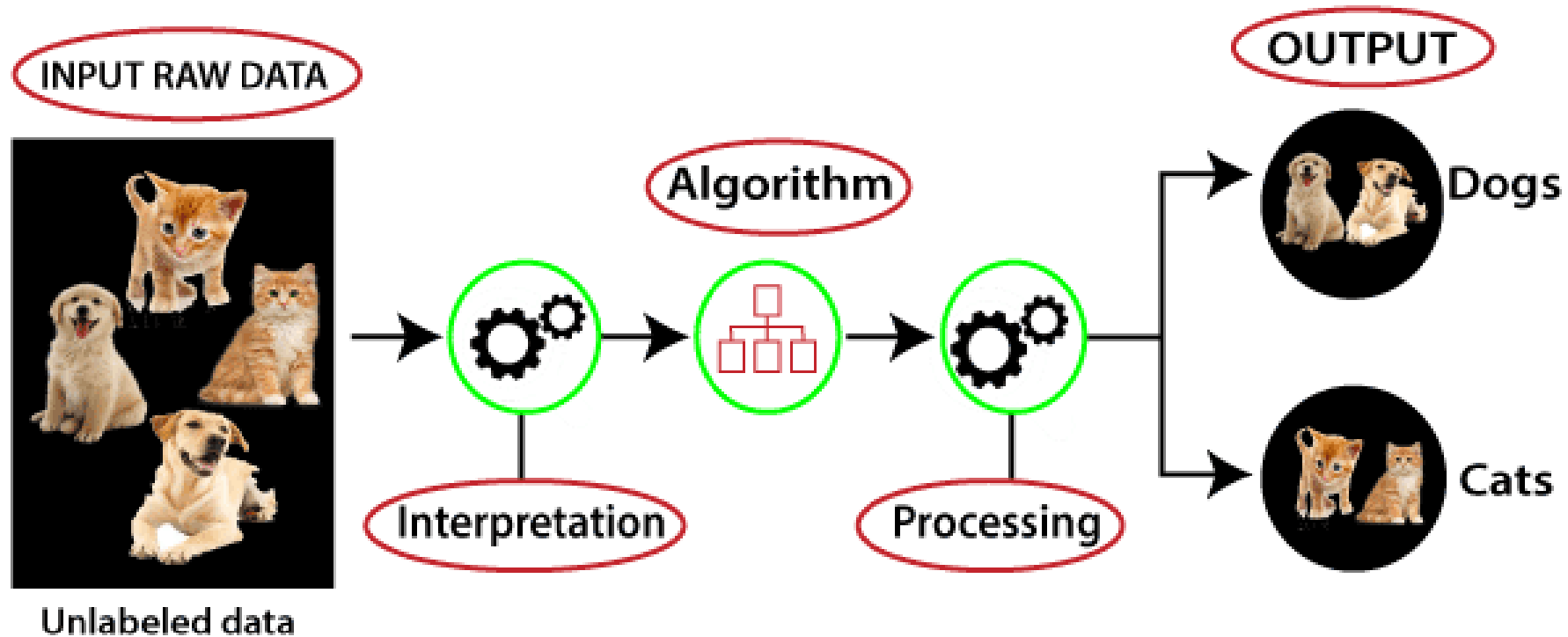
Unsupervised Machine Learning

Example: suppose the unsupervised learning algorithm is given an input dataset containing images of different types of cats and dogs. The algorithm is never trained upon the given dataset, which means it does not have any idea about the features of the dataset. The task of the unsupervised learning algorithm is to identify the image features on their own. Unsupervised learning algorithm will perform this task by clustering the image dataset into the groups according to similarities between images.





Working of Unsupervised Learning





Why use Unsupervised Learning?

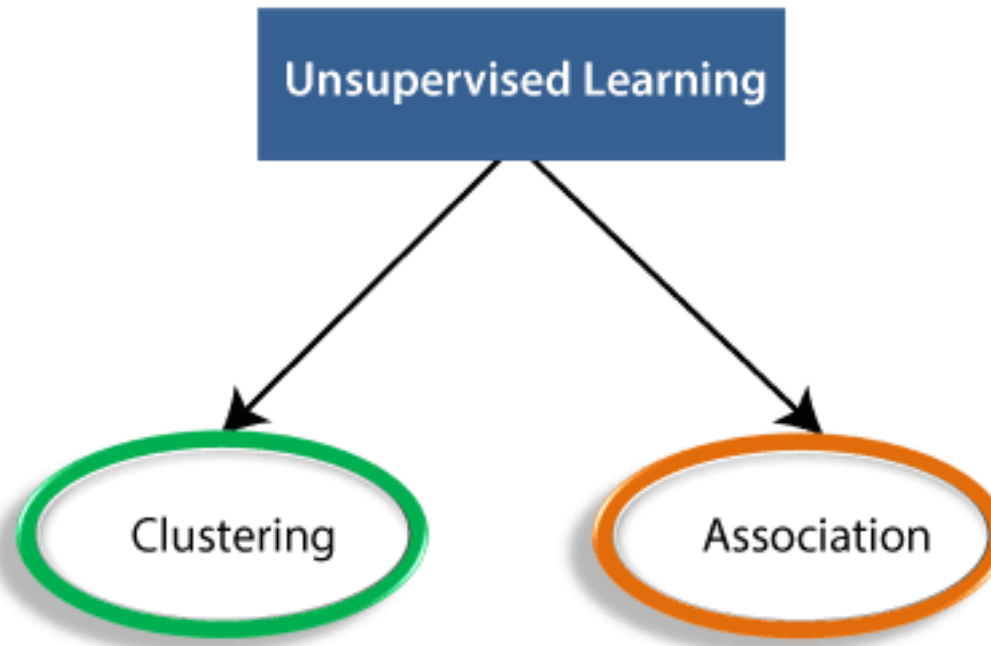
Below are some main reasons which describe the importance of unsupervised learning:

- Unsupervised learning is helpful for finding useful insights from the data.
- Unsupervised learning is much similar as a human learns to think by their own experiences, which makes it closer to the real ai.
- Unsupervised learning works on unlabeled and uncategorized data which make unsupervised learning more important.
- In real-world, we do not always have input data with the corresponding output so to solve such cases, we need unsupervised learning.



Types of Unsupervised Learning Algorithm:

The unsupervised learning algorithm can be further categorized into two types of problems:





Types of Unsupervised Learning Algorithm:

- **Clustering:** clustering is a method of grouping the objects into clusters such that objects with most similarities remains into a group and has less or no similarities with the objects of another group. Cluster analysis finds the commonalities between the data objects and categorizes them as per the presence and absence of those commonalities.
- **Association:** an association rule is an unsupervised learning method which is used for finding the relationships between variables in the large database. It determines the set of items that occurs together in the dataset. Association rule makes marketing strategy more effective. Such as people who buy X item (suppose a bread) are also tend to purchase Y (butter/jam) item. A typical example of association rule is market basket analysis.



Unsupervised Learning algorithms:

Below is the list of some popular unsupervised learning algorithms:

- **K-means clustering**
- **KNN (k-nearest neighbors)**
- **Hierarchical clustering**
- **Anomaly detection**
- **Neural networks**
- **Principle component analysis**
- **Independent component analysis**
- **Apriori algorithm**
- **Singular value decomposition**



Advantages of Unsupervised Learning

- Unsupervised learning is used for more complex tasks as compared to supervised learning because, in unsupervised learning, we don't have labeled input data.
- Unsupervised learning is preferable as it is easy to get unlabeled data in comparison to labeled data.



Disadvantages of Unsupervised Learning

- Unsupervised learning is intrinsically more difficult than supervised learning as it does not have corresponding output.
- The result of the unsupervised learning algorithm might be less accurate as input data is not labeled, and algorithms do not know the exact output in advance.



Supervised/Unsupervised Learning

Supervised Learning	Unsupervised Learning
Supervised learning algorithms are trained using labeled data.	Unsupervised learning algorithms are trained using unlabeled data.
Supervised learning model takes direct feedback to check if it is predicting correct output or not.	Unsupervised learning model does not take any feedback.
Supervised learning model predicts the output.	Unsupervised learning model finds the hidden patterns in data.
In supervised learning, input data is provided to the model along with the output.	In unsupervised learning, only input data is provided to the model.
The goal of supervised learning is to train the model so that it can predict the output when it is given new data.	The goal of unsupervised learning is to find the hidden patterns and useful insights from the unknown dataset.
Supervised learning needs supervision to train the model.	Unsupervised learning does not need any supervision to train the model.



Supervised/Unsupervised Learning

Supervised learning can be categorized in Classification and Regression problems.	Unsupervised Learning can be classified in Clustering and Associations problems.
Supervised learning can be used for those cases where we know the input as well as corresponding outputs.	Unsupervised learning can be used for those cases where we have only input data and no corresponding output data.
Supervised learning model produces an accurate result.	Unsupervised learning model may give less accurate result as compared to supervised learning.
Supervised learning is not close to true Artificial intelligence as in this, we first train the model for each data, and then only it can predict the correct output.	Unsupervised learning is more close to the true Artificial Intelligence as it learns similarly as a child learns daily routine things by his experiences.
It includes various algorithms such as Linear Regression, Logistic Regression, Support Vector Machine, Multi-class Classification, Decision tree, Bayesian Logic, etc.	It includes various algorithms such as Clustering, KNN, and Apriori algorithm.